



A new data security algorithm for the cloud computing based on genetics techniques and logical-mathematical functions

Fursan Thabit^{a,*}, Sharaf Alhomdy^b, Sudhir Jagtap^a

^a School of Computational Sciences, S.R.T.M. University, Nanded, India

^b Faculty of Computer and Information Technology (FCIT), Sana'a University, Yemen

ARTICLE INFO

Keywords:

Cloud computing security
Cryptography
Genetics techniques
Logical-mathematical functions

ABSTRACT

With the rapid development of distributed system technologies, one of the biggest challenges facing the digital world is ensuring the security of sensitive and confidential data during transport and storage, which are considered the most critical challenges facing cloud computing. There are many techniques to enhance data security on cloud computing storage environment. Encryption is the most significant method for data protection. Therefore, many available encryption algorithms are used to provide security, integrity, and authorized access using many methods such as DNA. However, they still have some limitations. In this paper, a novel variant of cryptography techniques is designed to improve cloud computing security by using two layers of encryption. The first layer is inspired by Shannon's theory of diffusion and confusion by the involvement of logical operations, such as (XOR, XNOR, shifting) with split the original plaintext and key into equal parts. The second layer is inspired from structures of genetics based on the Central Dogma of Molecular Biology for cryptographic purpose through simulation of the natural processes of genetic cryptography (translation from binary to DNA bases), transcription (regeneration from DNA to mRNA), and translation (regeneration from mRNA to protein). The experimental results enhanced data security that can be used to secure applications on cloud computing. The proposed algorithm's experimental results presented a strong security level, an apparent enhancement of cipher size and execution time compared to existing techniques widely used in cloud computing.

1. Introduction

Cloud computing is a new computing paradigm that provides various services on demand at a low cost. The central goal of cloud computing is to provide fast, easy-to-use computing and data storage services. Cloud computing is a modern computing model that offers low-cost, diverse resources on demand. Also, cloud computing's primary objective is to provide computing and data storage services that are fast, easy to use; in a cloud environment [1]. The computing world has excelled in dealing with cloud computing services, but some threats and risks arise in the cloud computing environment. For enhancing the security of cloud computing, there are many techniques such as cryptography which is the primary factor of security data in the cloud. It is achieved secrecy in the transfer of data or messages; it is also a way of converting the sender's letter into a secret term called the cypher text where only the intended recipient can learn the hidden message. Cryptology is a mathematical technique for security information management such as encryption, quality of data and authentication. Cryptography offers various more

robust methods, and it is possible to use techniques to provide these security services [2]. Encryption protocols, digital signatures, and hash functions various popular encryption algorithms are used to encrypt and decrypt confidential information that is represented by cryptography (Symmetric Algorithms, Asymmetric Algorithms, and Hybrid Algorithms) [3,4]. All these current schemes face protection issues. These schemes also take much time for cryptographic key generation, key retrieval, data encrypt and data decrypt.

Some of this researches are improved Deoxyribose Nucleic Acid (DNA) cryptographic with the requirements of DNA Cryptography systems, space of storage, speed, and security against attack in the digital world; through numerous studies and careful research, DNA cryptography has been developed whereby confidential data is hidden inside DNA molecules to enhance data security. DNA sequences are available as the National Center for Biotechnology Information (NCBI, DDBJ, and EBI) databases depending on the user data, the DNA sequence can be randomly selected, the biological process of DNA is challenging to break down, and it reduces the complexity of the calculation these factors

* Corresponding author.

E-mail address: thabtfursan@gmail.com (F. Thabit).

<https://doi.org/10.1016/j.ijin.2021.03.001>

Received 23 January 2021; Received in revised form 25 February 2021; Accepted 23 March 2021

Available online 24 April 2021

2666-6030/© 2021 The Author(s). Published by Elsevier B.V. on behalf of KeAi Communications Co., Ltd. This is an open access article under the CC BY-NC-ND

license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

Table 1

Requirement that fulfilled by the genetics techniques.

Requirement	Definition
1 Encrypt accomplished the full character set used GA encoding after convert it from binary to DNA	You must convert the entered text to the binary system and then replace it with a nucleic acid coding and encode for each the complete character set, which contains 96 items.
2 Generation Dynamic Table	The encryption schedule is randomly generated after each session The time interval to provide a different DNA sequence for each element Character set
3 Unique sequence Encrypt each character of the explicit transcript for DNA sequencing	Encryption the plain text in a DNA sequence is unique to each character set element in each generation of the coding table in each session.
4 Must increase the complex of genetics Cryptographic process	(Translation from binary to DNA bases), transcription (regeneration from DNA to miRNA), and translation (regeneration from miRNA to protein).
5 Simulate biological processes	The DNA coding and decoding algorithm must be based on a simulated biological process to adapt to the digital computing environment.
6 Dynamic encryption process	Plain script itself can produce a different coding script for each session due to the generation of the unique DNA coding table for each session.

Table 2

Key combination as Gen sequence in binary.

	A	T	C	G	T	G	C	A		
A	AA	AT	AC	AG	AT	AG	AC	AA	K1	SK1
T	TA	TT	TC	TG	TT	TG	TC	TA		
C	CA	CT	CC	CG	CT	CG	CC	CA	K2	
G	GA	GT	GC	GG	GT	GG	GC	GA		
T	TA	TT	TC	TG	TT	TG	TC	TA	K3	SK2
G	GA	GT	GC	GG	GT	GG	GC	GA		
C	CA	CT	CC	CG	CT	CG	CC	CA	K4	
A	AA	AT	AC	AG	AT	AG	AC	AA		

illustrate the need to encode DNA in cloud computing as the rapid calculation and increased complexity of cryptographic analysis were achieved. Some researchers evaluated the six characteristics of the active for DNA security measures identified in the standards encryption system [5].

The importance of DNA molecules has been proven by mathematical maximization Speed; reduce computational power and efficient storage.

In additionally advances in computing and cryptographic analysis want the development or enhancement of existing cryptographic systems. Researchers are attempting to develop new cryptographic technologies that can counter the threats posed by quantum computers. Recently; hurriedly growth bio-molecular computation so that bio-molecular computation applies in the cryptographic field and genetic cryptography is a new prototype cryptographic. A genetic algorithm is an

optimization technique based on Darwin's theory of natural selection. The genetic algorithm repeatedly modifies the set of individual solutions. The central population is established depending on the problem under study, and the suitability of each group is evaluated using the fitness function [6,7].

In addition, one of the biggest challenges facing cloud computing is ensuring the security of sensitive and confidential data during transport and storage because different attacks may occur. In this case, the encryption becomes more difficult if many and more complex keys are used.

The proposed research adopted the principles of a genetic algorithm to create robust subkeys. A gene-based homologous key coding scheme (DNA, RNA, and mRNA) with mathematical, logical function is proposed. The principle of genetic algorithm is a promising area that has attracted researchers in recent years. Deoxyribose Nucleic Acid (DNA) includes proposed unique genetic code made of monomers called deoxyribose nucleotides. Each DNA molecule contains two long strands of nucleotides, and each nucleotide is made of the deoxyribose sugar, phosphate group, and a nitrogen base. The nitrogenous bases are A (Adenine), G (Guanine), C (Cytosine), and T (Thymine) bonds with T and G bonds with C. These nucleotides appear in random order in each DNA molecule, and the arrangement of these nucleotides determines the genetic code of the DNA. The strength of any genetic scheme is directly related to the randomness it offers. The researchers explained that the randomness is in the nucleotide sequence could be efficiently applied to create robust and reliable coding schemes. The use of real DNA to improve coding systems is not currently potential due to complex laboratory requirements [8].

Moreover, to enhanced security issues in the cloud, this paper introduces a novel variant of cryptography techniques to improve the security of cloud computing. It has adopted two levels of encryption: the first level used the logical-mathematical function such as (XOR, XNOR, shifting) with split the original plaintext and key into equal parts and the second level used the genetics technique based on the Central Dogma of Molecular Biology (CDMB) for cryptographic purpose through simulation of the natural processes of genetic coding. Therefore, the major contributions of this paper can be summarized as follows:

- 1) A novel 256-bit cryptography technique based on CDMB technique with logical-mathematical function is introduced in this paper.
- 2) A novel 256-bit secret key generation technique based on DNA sequence is introduced in this paper
- 3) Security analysis and experimental results of Purpose work have been explained in this paper to verify its efficiency over the existing techniques.

The paper is organized in 9 sections as follows: Section (1) is dedicated to present an introduction. Section (2) a related work regarding the research. The requirement to be fulfilled by the genetics encryption algorithm is explained in Section (3). The proposed Algorithm has been presented in Section (4). Section (5) shows the implementation. The results and discussion are delineated in Section (6). The performance analysis is displayed in Section (7). Section (8) describes the security analysis and CIA achievements. The conclusions and further works are discussed in Section (9).

Table 3

Key combination as Gen sequence in binary.

	A	T	C	G	T	G	C	A		
A	0000	0001	0011	0010	0011	0010	0011	0000	K1	SK1
T	0100	0101	0111	0110	0101	0110	0111	0100		
C	1100	1101	1111	1110	1101	1110	1111	1100	K2	
G	1000	1001	1011	1010	1001	1010	1011	1000		
T	0100	0101	0111	0110	0101	0110	0111	0100	K3	SK2
G	1000	1001	1011	1010	1001	1010	1011	1000		
C	1100	1101	1111	1110	1101	1110	1111	1100	K4	
A	0000	0001	0011	0010	0011	0010	0011	0000		

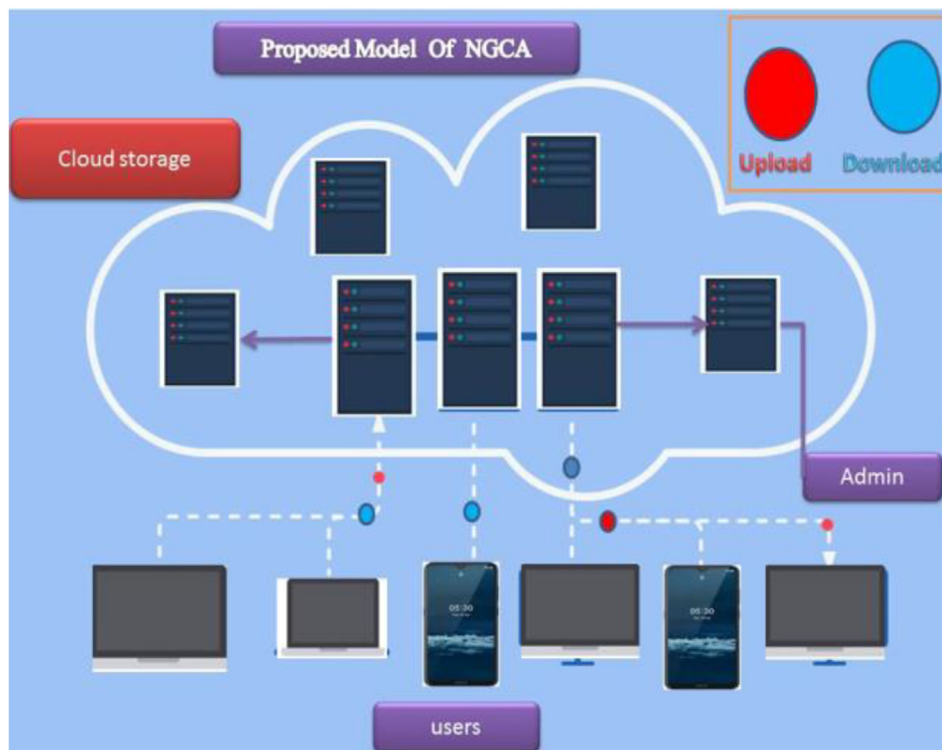


Fig. 1. The structure and conceptual workflow of the proposed model.

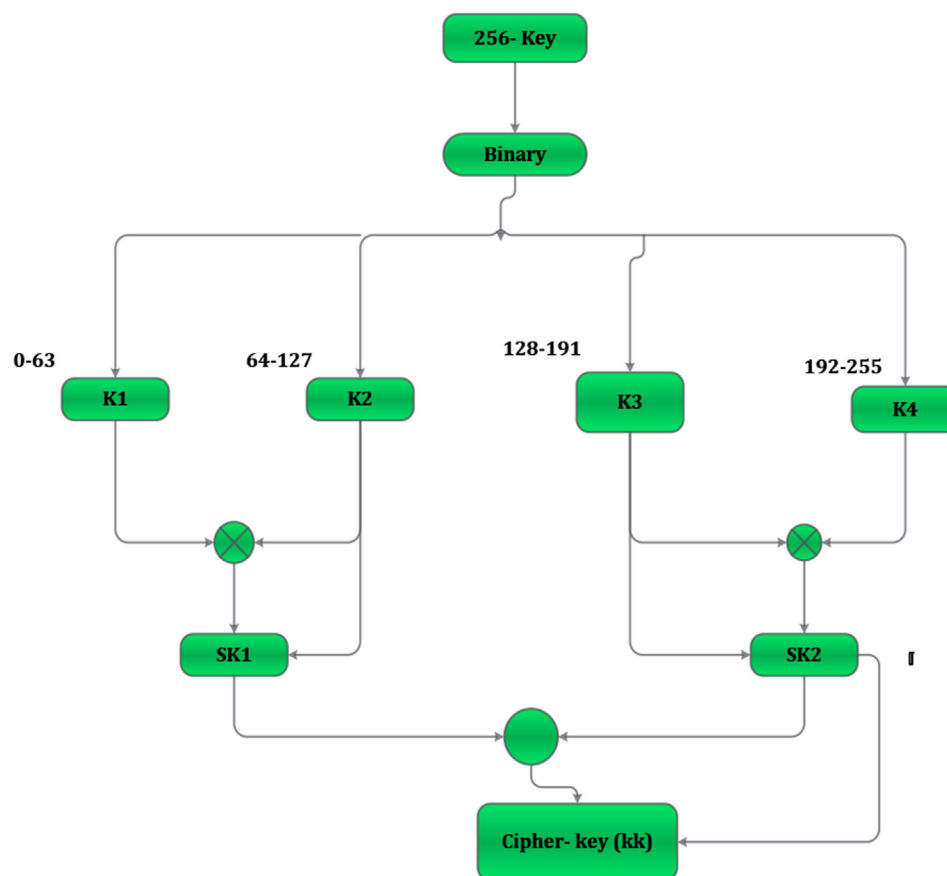


Fig. 2. Key generation process.

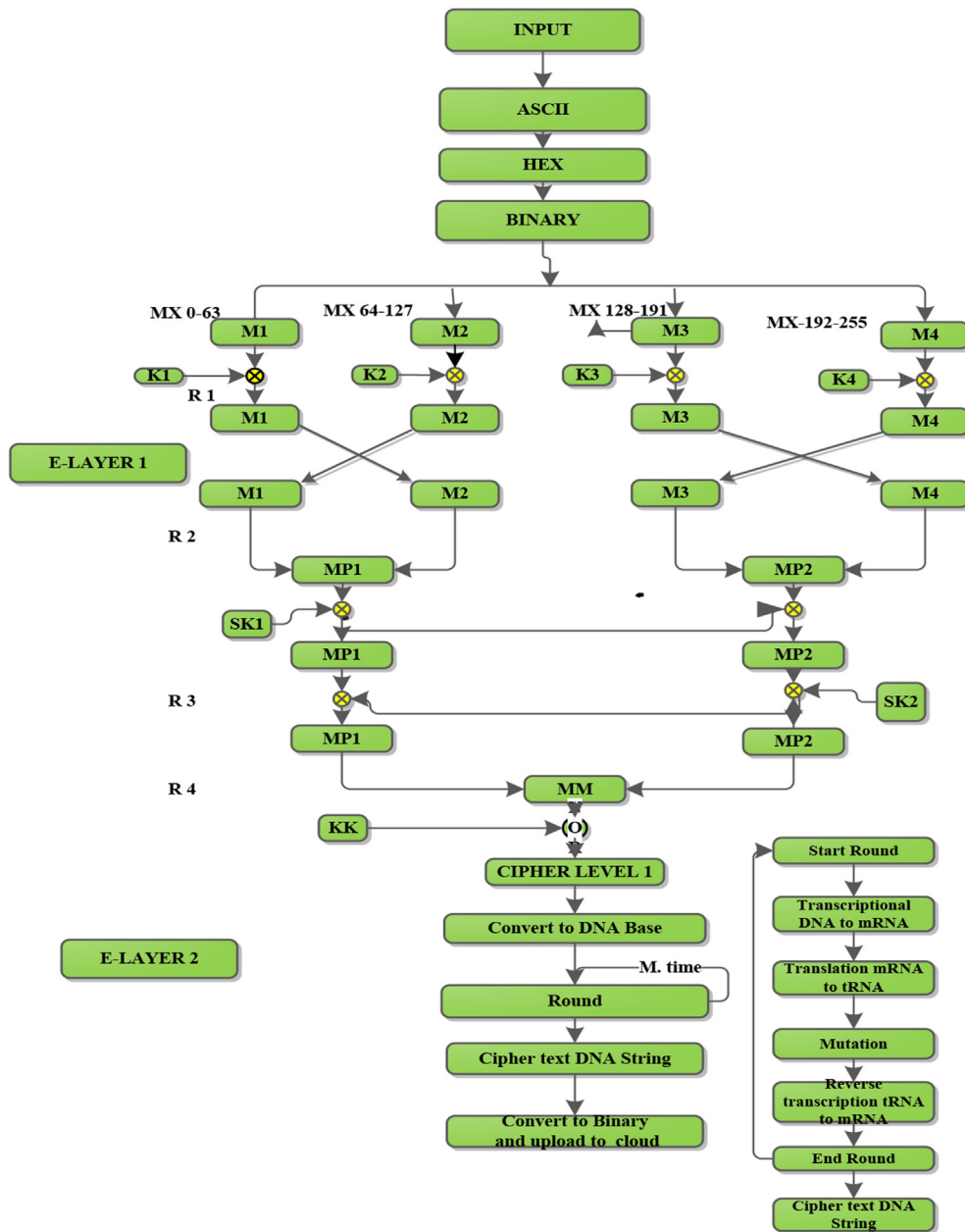


Fig. 3. Encryption process steps.

2. Literature review

Since its introduction to the market in 2000, cloud computing has been an active field of study. Cloud computing offers services such as virtual data storage, collaboration, servers, networks, applications and tools with minimal resource commitment. The central concern of cloud computing is the protection of information due to the vast volume of data stored in the cloud and communication systems' development, meaning that data can be hacked or destroyed from unauthorized access [9]. Cloud computing faces various security issues, some of which are service disruption, DoS attack, compromised authentication, and outside malicious insiders, security threats, system vulnerabilities, multi-tenancy issues, data integrity and data privacy [10]. Cloud technology guarantees the protection of data and handles cryptographic keys prevents data compromise or data failure. As there are many malicious users in the cloud, data protection could be at risk; cloud Service also has to maintain data security to prevent unwanted access to data in cloud storage. Incidents of data breaches or lack of data are widespread these days [11]

This authors analyzed data protection and privacy issues in cloud computing by concentrating on data isolation and secure privacy. Data security problems are of paramount significance to the extent and data of IaaS, PaaS and SaaS, which are core challenges in cloud computing. Data integrity is a critical enabler for cloud users. In addition to storing data, cloud technology typically offers data management facilities. In data integrity, Many scholars have also done a lot of work on cloud data integrity and privacy issues, some of which have been discussed [12]. Encryption is the key technology that satisfies the criteria of security [13]. Various algorithms based on symmetric/asymmetric core technologies and genetic techniques such as RSA, DES, etc. were adopted, developed and applied [14]. Cryptographic algorithms can be contrasted based on design, versatility, scalability, limitations and security, execution time, and memory requirements [15]. introduced the use of hybrid encryption approaches to enhance the privacy of the information stored on cloud servers such: Blowfish, RSA, AES, Eclipse IDA, DSA, This study focuses on not using third parties to encrypt customer data, but on allowing consumers the authority to decide ways of encrypting their

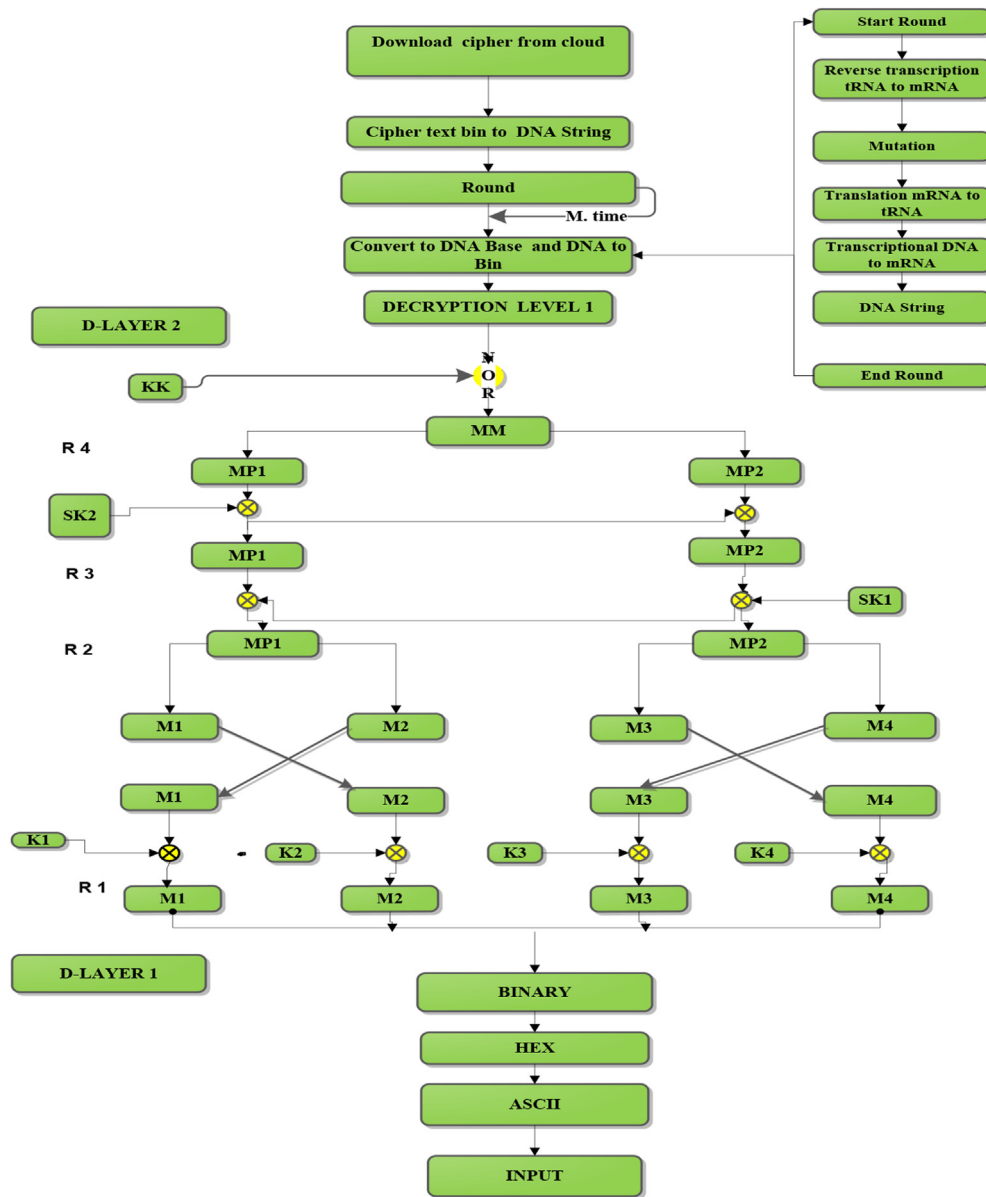


Fig. 4. Decryption process steps.

data" [16]. The authors identified previous studies devoted to cloud data security and performed a survey. They suggested a hybrid protection encryption approach using Blowfish and MD5 to offer improved security on the cloud server [17]. A comparison of IDAs, SHA-512, 3DES, and AES-256 made by Ref. [18], consists of on premise data encoding and decoding. This algorithm achieves a much higher level of protection and performance for large and small data files. This author [19] proposed a New Lightweight Cryptographic Algorithm for Enhancing Data Security that can be used to secure applications on cloud computing. The algorithm is a 16 bytes (128-bit) block cipher and wants 16 bytes (128-bit) key to encrypt the data. The authors [20] proposed a feature-based hierarchical encryption technique using semantic ontology for public auditing on cloud computing. The authors [21] exposed a GA/DNA hybrid model for encrypted images using a nonlinear shift reaction registry (LFSR) to create pseudorandom sequences. The authors [22] have proposed a GA-based approach to symmetric key generation to resolve the primary key distribution. The authors [23] introduced the computational cryptography of DNA combined with the in-depth learning of biological processes such as transcription, translation and genome sequencing. In Ref. [24] the authors proposed a symmetrical DNA-based

protection regime that processes binary data in the form of DNA. This technique uses an encrypted 128-bit or 64 nucleotide block scale. This coding scheme involves 16-round Feistel architecture, i.e. used in DES and AES; it provides random DNA to the algorithm but does not apply in real applications.

The authors [25] also implemented GA-based network security encryption solutions that guarantee the authentication, secrecy, non-repudiation and integrity of network messages being transferred. A new framework based on the Crypto-GA Genetic Algorithm (GA) is implemented in this analysis to fix data security and privacy concerns. GA is used to produce encryption and decryption keys Built with an encryption algorithm to ensure the protection and integrity of cloud data [26]. The DNA encryption method suggested by Aich et al. [27] uses Diffie-Hellman technologies to produce and exchange a hidden key that is translated to a DNA sequence. The simple text is translated to DNA sequencing, appended to prefixes, then converted to binary hybridization, and DNA hybridization is done using the transcript forming primary coding. The coding table is fixed here. Author [28] A novel DNA The suggested data encryption system for the cloud computing setting adopted in this article. The DNA bases and the complementary rule are

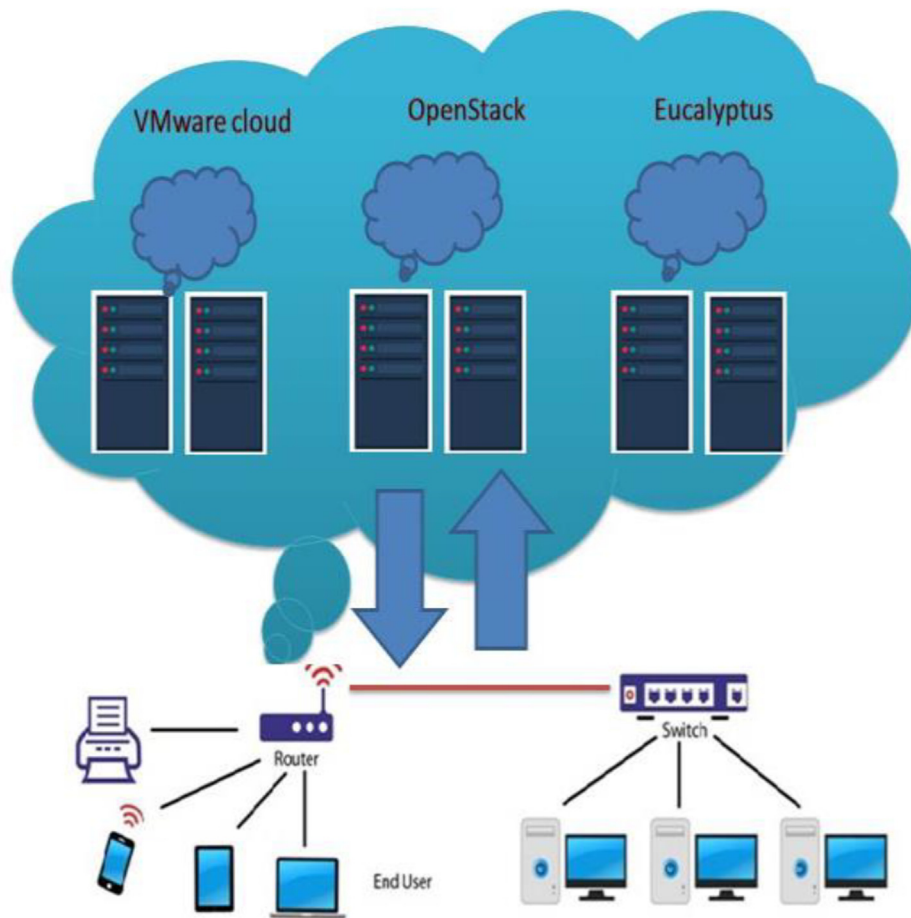


Fig. 5. Experimental steps diagram.

used to produce a hidden key that helps the device defend against many security threats. In this article, a novel and unique biological simulation technique focused on DNA encryption, and decryption has been established that fulfils both the functional and non-functional qualities that should be characteristic of a DNA computing-based encryption algorithm [29]. From the above-studied DNA encryption systems, they are very complex. It is evident that very few cypher schemes are dynamic and operate with static keys, coding tables, codebooks, etc. Using biological properties of DNA is much less among the proposed less secure cloud cipher systems data security.

3. The requirement that fulfilled by the genetics techniques for cryptographic algorithm

Each Cryptographic Genetics algorithm must satisfy a series of specifications. These criteria were established in this paper on the basis of the shortcomings found in the current cryptography algorithms as seen in Table 1.

4. The proposed algorithm

This section describes the proposed algorithm, which is a novel variant of cryptography techniques to improve the security of cloud computing by using two layers of encryption. The first layer is inspired by Feistel and substitution permutation with Shannon's theory of diffusion and confusion by the involvement of logical operations, such as (XOR, XNOR, shifting) with split the original plaintext and key into equal parts. The second layer is inspired by structures of genetics techniques based on the CDMB for cryptographic purpose through simulation of the natural

processes of genetic coding (translation from binary to DNA bases), transcription (regeneration from DNA to mRNA), and translation (regeneration from mRNA to protein). It also overcomes more extensive permutations without compromising Data security in the cloud.

4.1. Description

The main idea of the propose algorithm is to use the CDMB of Molecular Biology for cryptographic purposes. Because genetics has been used extensively to solve optimization problems with or without restrictions applied, genetics science is used in the natural sciences, mathematics, and widely in computer science. In computer science, genetics is used for both restrictive and unrestricted security and optimization problems; genetics science reduces enormous computational complexity by solving optimization problems in the least amount of time as it can solve difficult NP problems [26].

Genetics science is a bio-inspired calculation that modifies the individual solution of a repeatedly selected population; the primary processes of genetics are ("Population generation, crossover, and mutations"). Genetics science maximizes security as the technology structure differs from traditional security algorithms due to the used CDMB with the mathematical method. The proposed algorithm is more secure and best choice for the secure cloud environment. The structure and conceptual workflow of the proposed algorithm are shown in Fig. 1.

The conceptual illustration consists of two processes, i.e. upload and download data to and from cloud storage; the first input data is encrypted using the first layer encryption by logical operations, such as (XOR, XNOR, shifting) with the original plain text being divided into equal parts and then the process of merging each part with the primary and sub-keys

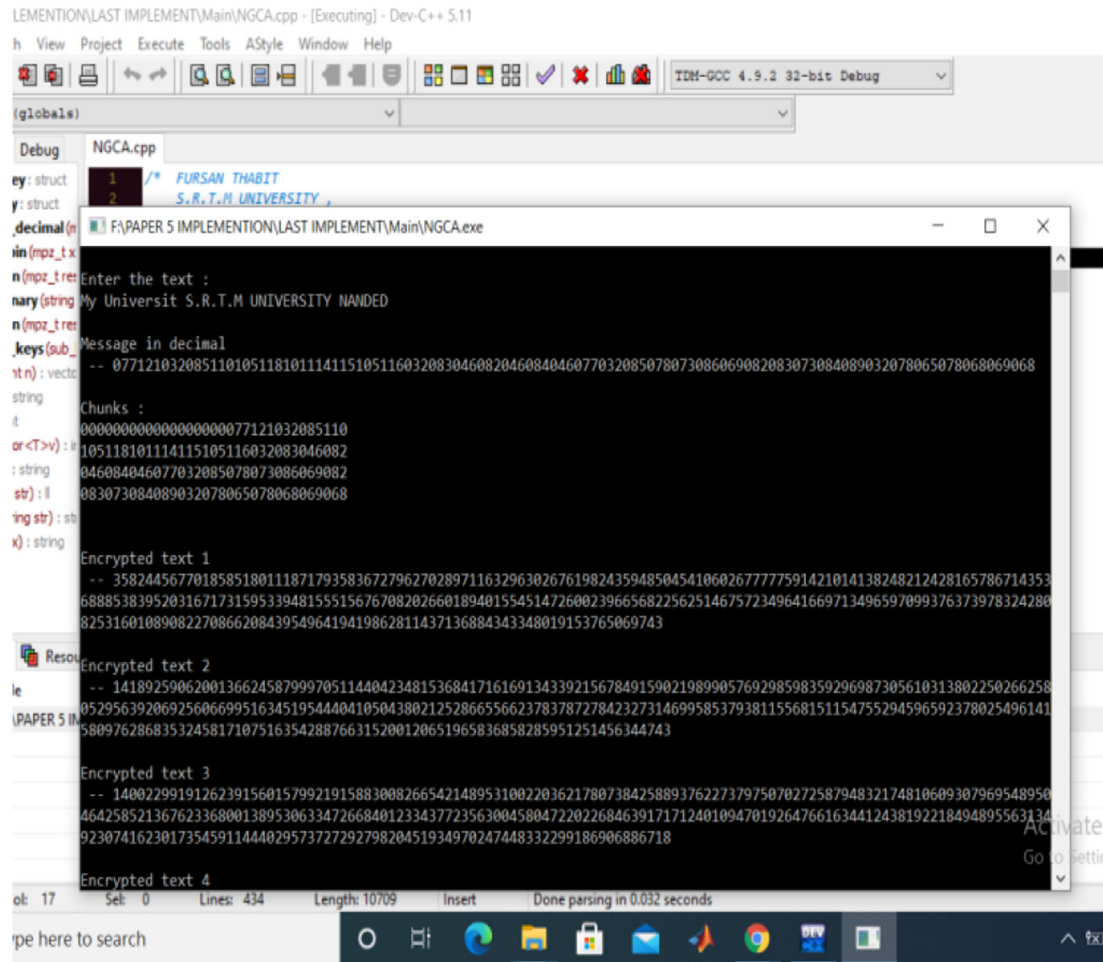


Fig. 6. Encryption process simulation.

that were generated in the key creation process, as shown in Fig. 2.

The second layer coding is performed to convert the output of the first layer coding to genetics-based (CDBM) the central doctrine of molecular biology by stimulating the natural processes of genetic coding translation from (binary bases to DNA bases from DNA to mRNA. mRNA to protein). A similar approach is reversed while downloading the data from cloud storage. The cryptography method key generation process, data encryption, and decryption are described next.

The proposed encryption algorithm consists of three stages.

1. Key Generation Process
2. Encryption Process
 - ✓ First Layer
 - ✓ Second Layer
3. Decrypting Process

4.1.1. Key formation process

- A. Select the two gene sequences, for example, ATGC and TCGA,
- B. The key generation process is as shown in Tables 2 and 3
- C. convert the key generation to binary
- D. The key is divided into four equal parts (K1, K2, K3, and K4).
- E. Used logical operations (XOR, XNOR) XOR between every two keys to generate Sub keys (Sk1, Sk2) and XNOR between two sub keys to generate the main key (KK) as shown in equations (1)–(3).

$$SK1 = K1 \oplus K2 + K2 \quad (1)$$

$$SK2 = K3 \oplus K4 + K3 \quad (2)$$

$$KK1 = SK1(xnor)SK2 + SK2 \quad (3)$$

The key is transferring to the receiver side by using Diffie-Hellman key sharing algorithm. For that, each time the key value will be randomly changed.

4.2. Encryption process

The encryption involved in this method is client-side, owner-side encryption, in which the data owner encrypts the data before uploading it to the cloud. After producing the keys (K1, K2, K3, K4, SK1, SK2, KK) from the key generation Process, the encryption process is performed to the input need for encrypting as in Fig. 2.

4.2.1. Encryption process steps

The encrypted message is separated into blocks in the encryption process steps (each referred to as M ... N) of equal length as shown in Fig. 3. The proposed algorithm will encrypt the block size 256-bit with 256-bit key for encryption. The next subsections will provide a description of the encryption process steps.

4.2.1.1. Encryption layer one_Logical-Mathematical functions.

- 1) Read the byte values from the input file known as plaintext and convert each character into its corresponding 3 –digit ASCII value.

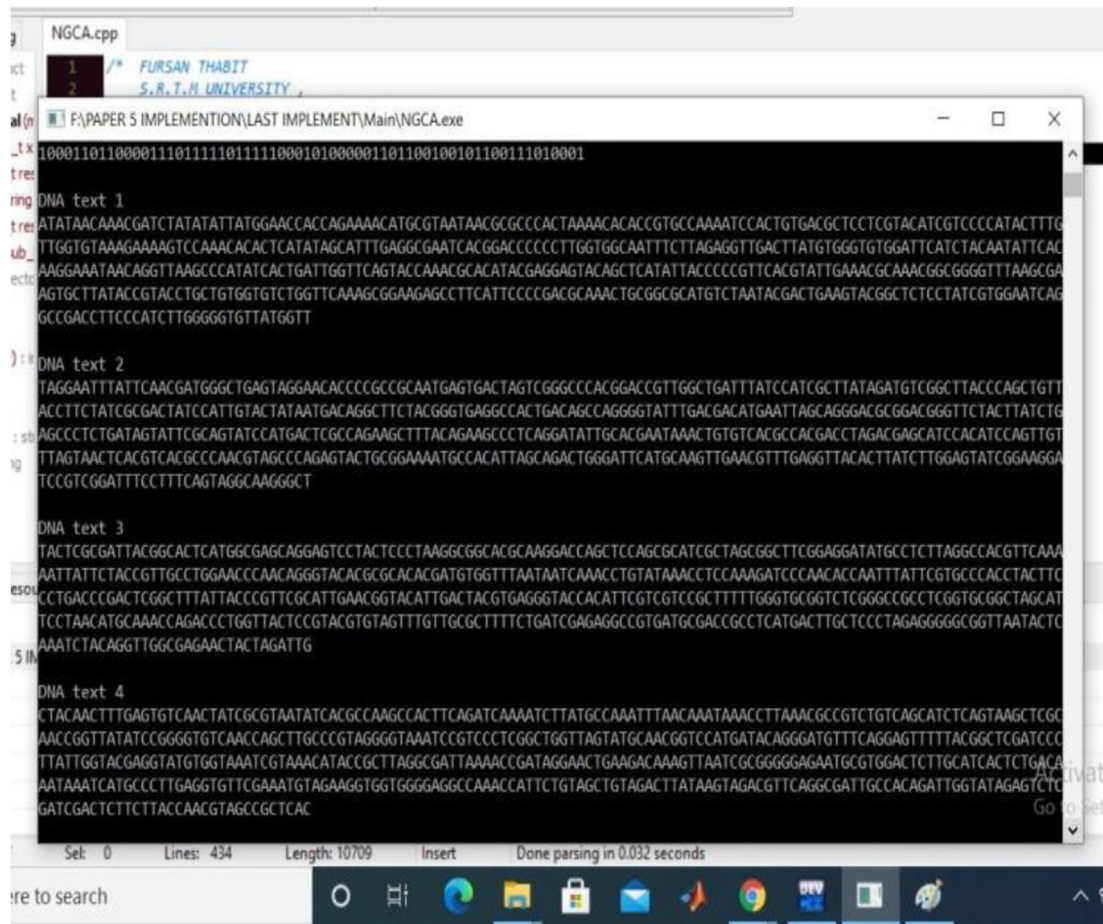


Fig. 7. Encryption process simulation.

- 2) Converts each integer digit into its' corresponding Hexadecimal value.
- 3) Transforms each hexadecimal value digit into its' corresponding binary value.
- 4) The binary value block is split into four 64-bit sub-blocks each, namely M1, M2, M3, and M4.
- 5) R1: Round 1 perform XOR operations between key and sub block as seen in Tables 2 and 3 and Fig. 2.
 $(M1 \oplus K1, M2 \oplus K2, M3 \oplus K3, M4 \oplus K4)$
- 6) R2: Round 2 its output the XOR operations
 $(M1 \oplus K1, M2 \oplus K2, M3 \oplus K3, M4 \oplus K4)$

Then swapping process takes place during the encryption process between the parts (m1, m2) and (m3, m4). As seen in Fig. 3.

- 7) R3: Round 3 combine (m1 + m2 to mp1 and m3 + m4 to mp2)
- 8) Perform XOR operations ($sk1 \oplus mp1$),
- 9) Perform XOR operations between output of the $sk1 \oplus mp1 \oplus mp2$.
- 10) Perform XOR operations ($sk2 \oplus mp2$),
- 11) Perform XOR operations between outputs of the ($sk2 \oplus mp2 \oplus mp1$).
- 12) Combine mp1 + mp2.
- 13) Perform XNOR operations between (KK) with (mp1 + mp2) to produce the cipher level 1.

4.2.1.2. Layer two_ genetics techniques.

- 1) Converts the binary value (cipher level 1) into DNA bases as 'A' = 00, 'T' = 01, 'G' = 10 and 'C' = 11.
- 2) "Transcriptional" DNA sequence is converted into an mRNA sequence by replacing (T) Thymine with (U) Uracil. This method is a simulation of a "biological" transcription process.
- 3) "Translation" The mRNA sequence is converted into a tRNA sequence by replaces each alphabet of DNA with its own complementary DNA alphabet. For example, A-U, U-A, G-C, and C-G transformations are performed. This method is a biological simulation of the translation process.
- 4) "Reverse Transcription" The tRNA sequence is transformed into DNA sequence by replacing uracil (U) with thymine (T) in the RNA sequence. This method is a simulation of biological reverse transcription.
- 5) Convert cipher text to Binary upload to cloud

4.3. Decryption process

The decryption process is reversed of the encryption process. It takes place after the user requests data from the cloud and confirms the user's login via the authentication process, as shown in Fig. 4. It consists of two stages. The first one, decrypt the second layer used the genetics technique based on the CDMB and the first layer decrypts the logical-mathematical


```

NGCA.cpp
1  /* FURSAN THABIT
2  S.R.T.M UNIVERSITY ,

F:\PAPER 5 IMPLEMENTATION\LAST IMPLEMENT\Main\NGCA.exe

338768745963295227947274899717315009033244104950345662172346818409683409

Decrypted text 1
-- 77121032085110

Decrypted text 2
-- 105118101114115105116032083046082

Decrypted text 3
-- 46084046077032085078073086069082

Decrypted text 4
-- 83073084089032078065078068069068

Decrypted text 1
00000000000000000000000077121032085110

Decrypted text 2
105118101114115105116032083046082

Decrypted text 3
046084046077032085078073086069082

Decrypted text 4
083073084089032078065078068069068

Original message :
My Universit S.R.T.M UNIVERSITY NANDED

Sel: 0 Lines: 434 Length: 10709 Insert Done parsing in 0.032 seconds
to search

```

Fig. 8. Decryption process simulation.

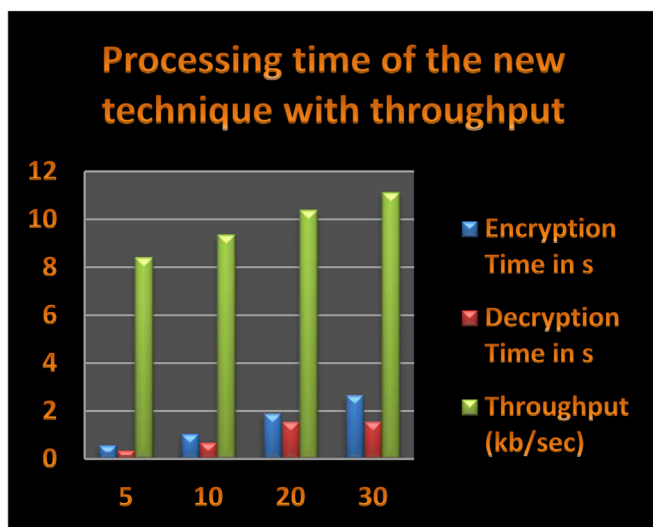


Fig. 9. Processing time of the new technique with throughput.

function such as (XOR, XNOR, shifting) with split the original plaintext and key into equal parts, as shown in Fig. 4.

4.4. Encryption detailed example

This section describes the example for proposed encryption algorithm

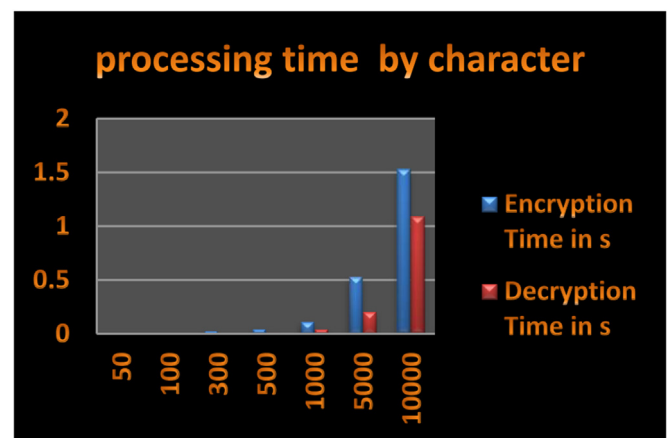


Fig. 10. Processing time of the new technique by character input.

in two encryption layers:

The first layer adopts the logical-mathematical.

Let us assume that the plaintext is "SRTM UNI" as follows:

- 1) Convert each character into its corresponding 3 –digit ASCII value (83 82 84 77 32 85 78 73).
- 2) Converts each integer digit into its' corresponding Hexadecimal value (53 52 54 4D 20 55 4E 49)

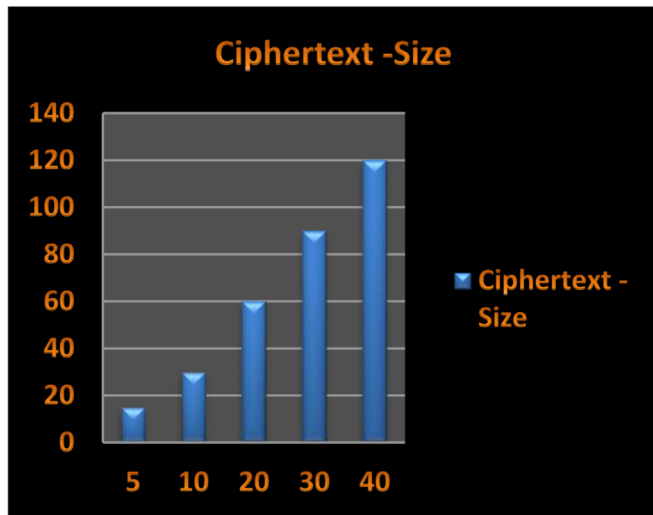


Fig. 11. Length Cipher-text of the new technique.

Table 4

Processing time of the new technique with throughput.

Plaintext Size (KB)	Encryption Time	Decryption Time (S)	Throughput (Kb/Sec)
5	0.59	0.381	8.41
10	1.06	0.712	9.35
20	1.92	1.581	10.39
30	2.69	1.594	11.1
40	3.59	2.59	11.1
Average	1.974	1.371	10.08

Table 5

Variable count characters.

Character count	Encryption Time (S)	Decryption Time (S)
50	0.0051	0.00204
100	0.0092	0.00328
300	0.030	0.0118
500	0.046	0.0208
1000	0.116	0.0446
5000	0.537	0.208
10000	1.53	1.09

Table 6

Length cipher-text.

Size in KB					
Plaintext Size	5	10	20	30	40
Cipher-text Size	15	30	60	90	120

Table 7

Processing time of the new technique with throughput.

Algorithm	Average Processing Time	Average Throughput(kb/s)
DNA(Goyal)	3.104	89.6
D-GET	3.33	20.92
BDEA	1.48	180.4
AES	1.61	126.8
DES	1.89	33.32
BLOWFISH	2.13	159.6
LED	2.9	22.92
PROPOSE -ALG	1.53	190.08

- 3) Transforms each hexadecimal value into its' corresponding binary value 01010011 01010010 01010100 01001101 00100000 01010101 01001110 01001001.
- 4) The block of the binary value is generally divided into four sub-blocks of 64 bits each, namely, M1, M2, M3, and M4. M1 = 01010011 01010010
M2 = 01010100 01001101,
M3 = 00100000 01010101,
M4 = 01001110 01001001.

Also the key is generated and divided according to the text size as in Fig. 2.

K1 = 00000001 00110010
K2 = 01000101 01110110
K3 = 11001101 11111110
K4 = 10001001 10111010
SK1=K1⊕K2+K2 = 01000100010001000100101 01110110
SK2=K3⊕K4+K3 = 010001000100010010001001 10111010
KK=SK1 XNOR SK2 + SK2 = 1111111111111111
00110000001100110100010001000100100010011011010.

- 5) Now in Round 1(R1): Perform XOR operations between key and sub block as seen in Tables 2 and 3 and Fig. 2. (M1⊕ K1, M2 ⊕ K2, M3 ⊕ K3 M4 ⊕ K4).

M1⊕ K1 = 0101001001100000
M2 ⊕ K2 = 0001000100111011
M3 ⊕ K3 = 1110110110101011
M4 ⊕ K4 = 1100011111110011

- 6) Round 2 (R2): Its output the XOR operations (M1⊕ K1, M2 ⊕ K2, M3 ⊕ K3 M4 ⊕ K4), with swapping process takes place during the encryption process between the parts (m1, m2) and (m3, m4). As seen in Fig. 3. M1 = 0001000100111011

M2 = 0101001001100000
M3 = 1100011111110011
M4 = 1110110110101011

- 7) Output the swapping feed process of round 2 (R2): combine (M1+ M2 to MP1 and M3 + M4 to MP2),

M1 = 0001000100111011
M2 = 0101001001100000
MP1 = M1+M2
00010001001110110101001001100000
M3 = 1100011111110011
M4 = 1110110110101011
MP2 = M3+M4 = 1100011111110011110110110101011

- 8) Perform XOR operations (SK1 ⊕ MP1) SK1 ⊕ MP = 10101010101111110001011100010110

Perform XOR operations between (SK1 ⊕ MP1) with MP2 = 1001001010001100111101010111011.

- 9) Perform XOR operations (SK2 ⊕ MP2) = 10000011101101110110010000010001.

Perform XOR operations between (SK2 ⊕ MP2) with MP1 = 10010010100011000011011001110001.

- 10) Combine output of MP1+ MP2.

P = 10000011101101110110010000010001100100101000
11000011011001110001

- 11) Perform XNOR operations between (KK) with (MP1+ MP2) to produce the cipher level.

P = 10000011101101110110010000010001100100101000
11000011011001110001
KK = 1111111111111111001100000011001101000100010
001001000100110111010
P = XNOR KK = 10000011101101111010101111
0111010010100100110111010000000110000.

- 12) Cipher text layer 1 was produced

10000011101101111010101110111010010
100100110111010000000110000.

The second layer adopts the genetics technique as follows:

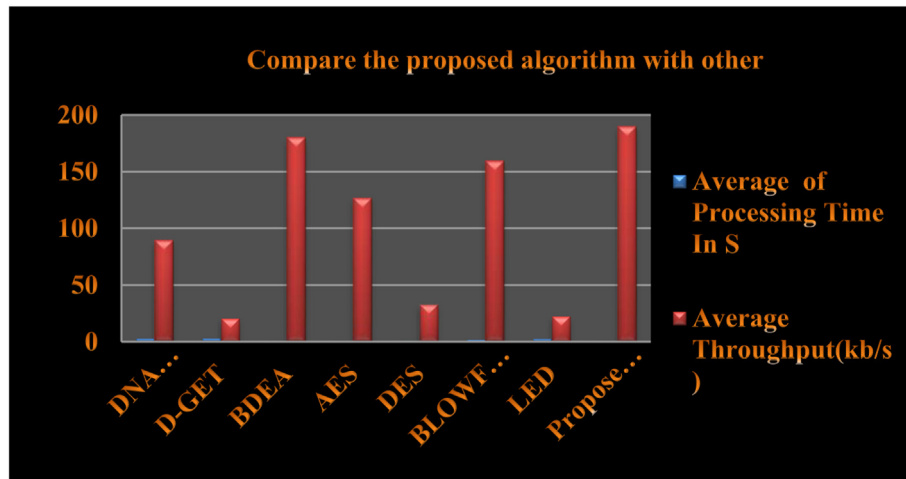


Fig. 12. Result of processing time with Throughput.

Table 8

Comparative analysis between the proposed technique and symmetric encryption techniques based on different parameter.

	DES [30]	AES [31]	BLOWFISH [32]	LED [33]	NLCA [19]	PROPOSED ALGORITHM
Structure	Feistel	Subs-Per	Festial	Festial	Festial/SP	Festial/SP/Genetics science
Key Size	64 bits	128 bit	64 bits	64 or 128	128, or 256	128/256/512/1024
Block Size	56 bits	128, 192, 256 bits	32–448 bits	64 or 128	128, or 256	128/256/512/1024
Possible Key	2^{56} bits	2^{128} , 2^{192} Or 2^{256} bits	2^{32} – 2^{448} bits	2^{64} , 2^{128} bits	2^{128} , 2^{256} bits	2^{128} , 2^{256} , 2^{512} , 2^{1024} bits
Deposit Of Keys	no need	needed	needed	needed	needed	needed
No. Of Round	16	10, 12, 14	16	Variable	4	layer 1 = 4 layer 2 = 3
Mathematical Operations	Permutation, Xor, Shifting, Substitution. (6 bits input 4 output bits)	Xor, Mixing, Substitution, Shifting, Multiplication, Addition. (16 bits)	Xor, Mixing, Substitution, Shifting, the S-boxes accept 8-bit input and produce 32-bit output	XOR, Rotations, 2n Mod Addition, Substitution (6 bits)	Xor, Xnor, Shifting, Substitution (4 bits)	Layer1(Splitting, XOR, XNOR, Shifting, Substitution Swapping) Layer 2 (Transformation-Translation) Faster
Encryption Process	Slower	Moderate	Moderate	Fast	Faster	Faster
Decryption Process	slower	Moderate	Moderate	Fast	Faster	Faster
Power Consumption	Very low	Low	Low	High	High	High
Security	Less Secure	Secure	Secure		Secure	High Secure

- 13) Converts the binary value (cipher layer 1) into DNA bases as 'A' = 00, 'T' = 01, 'G' = 10 and 'C' = 11,

10 00 00 11 10 11 01 11 10 10 11 11 01 11 01 00 10 10 01
00 11 01 11 01 00 00 00 00 11 00 00

GAACGCTCGGGCCTCTAGGTACTCTAAAACAA

- 14) "Transcriptional" DNA sequence is converted into an mRNA sequence by replacing (T) Thymine with (U) Uracil. This method is a simulation of a "biological" transcription process

GAACGCU CGGGCCUCUAGGUACUCUAAAAACA
A

- 15) "Translation" The mRNA sequence is converted into a tRNA sequence by replaces each alphabet of DNA with its own complementary DNA alphabet. For example, A-U, U-A, G-C, and C-G

GUUCGCA CGGGCCACAAGGAUCA CUUUUCU
U

- 16) "Translation" The mRNA sequence is converted into a Reverse tRNA sequence by replaces each alphabet of DNA with its own complementary DNA alphabet. For example, G-C, and C-G

CUUCGAGCCCGGAGAACCAUGAGUUUUUGU
U

- 17) Reverse transcription" The tRNA sequence is transformed into DNA sequence by replacing uracil (U) with thymine (T) in the RNA sequence. This method is a simulation of biological reverse

CTTGCGAGCCCGGAGAACCATGAGAAAAAGA
A

- 18) Convert cipher text to Binary Send over Public Channel.

CTTGCGAGCCCGGAGAACCATGAGAAAAAGA
A

- 19) Cipher-text = 11 01 01 10 11 10 00 10 11 11 11 10 10 00 10 00 00
11 11 00 01 10 00 10 00 00 00 00 00 10 00 00

- 20) Upload the cipher text to the cloud.

4.5. Decryption process example

The decryption process is reversed to the encryption process. It consists of two stages. The first stage is to decrypt the second layer of genetics technology based on CDMB the central tenet of molecular biology during the simulation. It is converted the coded text into the binary system and then starts the reverse simulation process (DNA to protein, protein to tRNA, tRNA to mRNA, mRNA, to DNA) and convert the DNA to binary base 2 bit. The second stage is decrypted the layer one logical-

Table 9

Analysis differentiates between the proposed technique and other genetics techniques.

	DNA-ELGAMAL [34]	RC4-DNA-ALG [35]	RSA-DNA-ALG [36]	CRYPTOGA [26]	GOYAL [37]	BDEA [38]	D-GET [7]	PROPOSED ALGORITHM
1 Overview	In this paper DNA cryptosystem is proposed to secure the original data within the DNA nucleotides providing greater storage space, reduced overhead and dynamic operations.	In this paper, DNA based algorithm which acts as an additional layer around the RSA algorithm. A round function based on the process of protein synthesis from DNA.	The algorithm implementation of some interesting characteristics of DNA sequences. with RSA Both of DNA's features and Binary Coding Technology beside Complementary Pairing Rules are explained through the paper	In this paper, a new model based on a genetic algorithm (GA) CryptoGA is proposed to cope with data integrity and privacy issues. GA is used to generate keys for encryption and decryption which are integrated with a cryptographic algorithm to ensure privacy and integrity of cloud data.	In this presented work the DNA based cryptographic algorithm for security is used. The DNA based cryptographic technique is basically developed using the substitution and other basic operator's implementation	Bio-Inspired Cryptosystems are a modern form of Cryptography where bio-inspired and machine learning techniques are used for the purpose of securing data. A system has been proposed based on the Central Dogma of Molecular Biology (CDMB) for the Encryption and Decryption Algorithms	In this paper, propose DNA-Genetic Encryption Technique (D-GET) in order to make the technique more secure and less predictable. In this technique, binaries any type of digital data and convert it to DNA sequencing, reshape, encrypt, crossover, mutate and then reshape. DNA	In this paper, a novel variant of cryptography techniques to improve the security of cloud computing by using two layer of encryption, first layer is inspired by Shannon's theory of diffusion and confusion. The second layer which is inspired from structures of Genetics techniques
2 layer	DNA- ElGamal	RC4-DNA	DNA + RSA	DNA	DNA	DNA	DNA	Genetics Techniques- Logical- Mathematical Functions
3 Architecture Structure	DNA nucleotides + ElGamal	DNA nucleotides with RSA	DNA with Binary Coding	Caesar Cipher + DNA	DNA nucleotides	CDMB	DNA nucleotides	DNA/mRNA/ tRNA/protein + Mathematical Functions
Key size	Randomly	Randomly	–	128	Randomly	Randomly	Randomly	64/128/256/ 512
Block size	104 bits	256 bit	–	128	256	–	–	64/128/256/ 512
No. of Round	1	no	no	no	no	no	no	4
Encryption/ decryption Algorithm	Fast	slow	slow	fast	slow	fast	fast	faster
Algorithm	Partially sy and asy	Asymmetric	symmetric	symmetric	symmetric	Partially Sy and Asy	symmetric	symmetric
4 Requirement Fulfilled for G T								
5 Complete character encoding			*	*	*	*	*	✓
5 Dynamic k encoding	x	x	x	yes	x	✓	*	✓
4 Unique sequence	✓	x	x	*	x	*	*	✓
5 Robustness	✓	*	*	✓	*	*	✓	✓
5 Biological process	*	✓	✓	✓	✓	✓	✓	✓
6 Dynamic encryption pro	*	x	x	✓	*	✓	*	✓
7 Flexibility Security & Limitations	yes secure	no secure	no Less secure	yes secure	no Less secure	yes secure	no Less secure	yes High secure

mathematical functions shown in Fig. 4.

5. Implementation

This section describes the experimental steps of the proposed algorithm on multi-cloud cluster. A multi-cloud comprised of three well-known private clouds, i.e. OpenStack, VMWare Cloud Suite, and Euca-lyptus. Fig. 5. Show the configuration of cloud architecture. A single node has been used to create a cloud group, such as a mutual belvedere between clouds.

- VMware vCloud Suite is cloud-ready for enterprises management platform. It is best for hybrid heterogeneous clouds. VMWare vCloud Suite supports monitoring software (Hyper-V and KVM) as ESXi hypervisor is recommended. ESXi is Hypervisor that builds and operates virtual machines if the interface is consistent with the host. VMWare consists of VCloud Director, VShield Admin, VCenter Server, and VChar modules.
- OpenStack is an open-source cloud computing platform. It has been developed by NASA and Rackspace Hosting. Clients like IaaS have access to a resource like VPS etc. A wide range of compatible vendor

infrastructure can be built. The client has a range of ways to handle it like a dashboard, command line, or network services. It comprises of Horizon (Dashboard, Swift (Object and File Storage System), Cinder (Block Storage-A file access mechanism), Nova (Computing Engine or VM Handler) components), At a Glance (Image Service-VM Templates), Neutron (Networking Mechanism), Ceiling Scale (Remote Billing Service), Heat (Synchronization - Cloud Service Infrastructure Requirements for Operation), and Keystone (Identity Services - Map Permission against Access).

- C. Eucalyptus: an open-source private cloud compatible with AWS *. Eucalyptus is open-source software for created AWS, QA and Cloud compliant proprietary IT technologies. Makes it easy to deliver cloud computing, just like AWS, from within your data center.

Emphasis has been placed on implementing the proposed algorithm that was conducted in a private - cloud using (Eucalyptus and VMware vCloud) running on an Intel Core™ i7-3120 M, 2.50 GHz, 16 GB processor RAM. Three clusters are used to create a group of clouds with sufficient numbers of n-nodes (owners/users) to represent each node; the node that starts uploading the file is the owner who encrypts the file and shares it with the user; the data owner encrypts the data and transfers it to the data user. The data owner decrypts the data in order to enter the data.

Algorithms have been implemented in MATLAB as well as implemented in C ++ language using dev C ++ program as shown in Figs. 6–8. The Various parameters have been evaluated, i.e. encryption/decryption execution time, throughput, length Cipher-text, security standard and key size and performance efficiency analysis are also conducted for comparison in the next section.

6. Results and discussion

To verify the proposed algorithm, several security experiments and analyzes are performed to assess the robustness of the proposed coding scheme based on the following the evaluation parameters; encryption time, decryption time, throughput and length cipher-text in section 6.1. Whereas in section 6.2 compared the proposed algorithm with other genetics encryption techniques and existing symmetric key encryption techniques such as AES, DNA, DES, and Blowfish, based on the size of plain text, the time required to encryption/decryption.

6.1. Evaluation parameters of the proposed algorithm

6.1.1. Encryption time

Encryption time can indeed be described as the time taken to convert plain text to cypher text. The encryption algorithm's quality is inverse relationship proportional to the time of the encoding; always when the technique takes less encryption time, it will be more efficient. This parameter measured in two ways:

- A. Analyze the encryption time based on the input size (10, 20, 30, 50, and 100 KB) as shown in Table 4 and Fig. 9.
- B. We analyze the encryption time base on input by variable count characters, time encryption and decryption is calculated, as shown in Table 5 and Fig. 10.

From the results obtained, it was shown that perhaps the encryption process time increases linearly with just an increase in the number of character or the size of the file, but the time taken to decipher process is less than the decryption process time. It demonstrates that the computational complexity is much less in proposed work, as shown in Figs. 10 and 11.

The encryption time to encrypt the plain text of the proposed algorithm less time than another genetic algorithm, the symmetrical algorithm AES, DES, and Blowfish. As shown in Table 7 below.

6.1.2. Decryption time

Algorithm Decrypting Time shows the time taken to recover the original data using cipher text entry. The amount of time it takes is also called the time complexity of the decrypting algorithm. The time spent decrypting can be calculated using the following formula.

Time consumed = end time-start time

According to the performance obtained as shown in Tables 4 and 5, the proposed technique takes less time to decode as compared to other encryption algorithms. Thus the proposed model is adoptable and effective for safe communication.

6.1.3. Throughput

The efficiency of an algorithm can be analyzed directly from its throughput. The throughput of the algorithm is directly proportional to the performance, that is, the higher the performance of the algorithm, the higher the throughput. The formula for calculating the throughput of the encryption technique is as following:

Throughput = Plain Text Size/Encoding Time

Based on the encoding time reported in Table 4. The throughput for all techniques was estimated using the formula above. Table 4 and Fig. 9. Clearly demonstrate that the throughput is better than any other techniques. Thus, it can be inferred that the proposed algorithm does better than other genetics cryptography techniques in and symmetric key algorithms.

6.1.4. Length cipher-text

Since then, this technology has been developed to store data in a Cloud file, so cloud storage has been taken into account when planning this method so that the scale of the data is not any greater after encryption. Besides, the transfer time is directly proportional to the sum of the data received. Since then, in client-side encryption, the data is encrypted before being uploaded to the server; the amount of data downloaded directly influences transmission time. We compared our solution and others, and the findings revealed that our system generates fewer coded text than the existing coding techniques. The results are seen in Table 6 and Fig. 11 below:

6.2. Comparison the proposed with the other techniques

In this section, comparative studies are conducted to view and check the feasibility of the proposed algorithm.

6.2.1. Compare the proposed algorithm with other

The first section was conducted, taking into account the time requirements for encoding and decoding processes, and the algorithm's productivity. Comparisons with some well-known cryptographic techniques such as DNA (Goyal), D-GET, BDEA, AES, DES, BLOWFISH and LED. As shown Table 7 in Fig. 12, respectively. The proposed method requires relatively less time than other techniques, and the throughput is efficient.

6.2.2. Comparative analysis between the proposed technique and symmetric encryption techniques

This section presented the Comparative analysis between symmetric encryption techniques which are commonly used for data security in cloud computing with the proposed approach. A brief comparison is shown in Table 8. Showing Structure, Key Size, block size, Possible key, Deposit of keys, No. of Round, Mathematical Operations, Encryption process, Decryption process, Power consumption, security, for the frequently used cryptographic schemes, namely DES [30], AES [31], BLOWFISH [32], LED [33], NLCA [19], together with the proposed algorithm.

6.2.3. Analysis differentiates between the proposed technique and other genetics techniques

In this part, the analysis distinguishes between the proposed approach and other genetics techniques which commonly used for data security in cloud computing, based on some well-known parameters used to evaluate the cryptography techniques and other evaluated parameters fulfilled by genetics encryption techniques. A brief comparison is shown in Table 9 with the frequently used cryptographic genetics schemes, such (DNA-ElGamal [34], RSA-DNA-Alg [35], DNA-RC4-Alg [36] CryptoGA [26] Goyal [37], BDEA [38], D-GET [7]) together with the proposed algorithm.

It can be seen from the tables that the proposed algorithm has high-security strength and is very flexible. It also required relatively less time than other techniques and the throughput is effective as shown in Tables 7–9. It is also noted that the proposed algorithm reduces computational complexity and reduces Processing strength complexity by having a clear structure consisting of two layers of encryption and containing only four rounds of coding each round requires only simple math with genetics simulation process.

7. Performance analysis

7.1. Confusion and diffusion

Confusion and diffusion are the security features Shannon first discusses [39]. Confusion indicates that each bit of the cipher-text should be based on several bits of the key. In other terms, the relation between the cipher-text and the key must be as complicated as possible. In the proposed cipher scheme, the cipher-text, converted to binary and used logical operations, such as (XOR, XNOR, and shift) with the original regular text and key divided into equal parts. In addition, the property is facilitated to some degree by CDMB. The translation is a one-to-many interaction, which includes simulating the natural processes of genetic coding (translation from binary bases to DNA bases), transcription (regeneration from DNA to mRNA) and translation (regeneration from mRNA to protein); In this complex cryptography technique, it is not possible to use plain text or the key to draw the cipher-text independently, as many operations are performed to combine each to obtain the result. Diffusion means that if one character of the standard text is altered, a few hierarchical text parts will be changed. This is done using the suggested coding scheme, where encoding and translating operations would lead to modifications in the cypher text as one section of the plain text is modified. It is also necessary to note that the same plain text will produce separate cypher scripts using the proposed cypher scheme since the generated key will be different each time. The suggested method thus achieves the features of confusion and diffusion.

7.2. Impact of the avalanche

Avalanche Effect is an encryption property where a small change in the standard text results in a substantial change in the cipher-text for the current encryption scheme, randomly changing 1 bit in plain text (in binary) resulted in changes to 92.65% bits of coding-text on average, thus changing 2 bits in plain text (in binary form) results randomly in changes to 97.11% bits of cipher text on average. It implies that the proposed coding scheme has a powerful randomization distribution. It also allows for complicated and complex mapping between the regular text and the cypher text.

8. Security analysis

This section analyzes the strength and the attacks for the cipher-text produced by the proposed technique.

8.1. Analyze the strength of the cipher text

The proposed technology includes two-layer of security during encrypted text generation, as follows.

- Layer 1: It investigates Shannon's theory of diffusion and disorientation by engaging logical processes, with feudal and substitution architectural methods to improve cryptography's complexity. To breach the data secret, the intruder must perform 2^{256} , 2^{512} , times a brute force attack on the key.
- Layer 2: When the data is distributed using DNA cryptography, the DNA used as data is transmitted either in the form of ATCG, which simulates the natural processes of genetics and related technologies. In order to violate confidentiality, the attacker must also replicate the procedure at the same time to produce dynamic DNA coding, assuming that the sender divides the data into N segments and that the amount of data in each part is P. Also assuming that it selects the mi DNA base sequence from the NCBI bank database to merge the first two segments. Now to breach its secrecy, the number of operations an intruder needs to perform is: $(N \times P \times mi)$. Due to the complexity of the two stages, the strength of the encryption is very high.

8.2. Attack analysis

The strength of any cryptographic procedure has to do with its sustainability against attacks. Usually, in order to infringe the secrecy of the records, the attacker attempts to initiate various potential modes of attack. Specifically, for cipher parsing, it tries with brute force attack, known plain text attack, cipher only text attack and selected plaintext attack besides manipulating or fabricating cipher, it tries to launch, noise attack, blockage attack (data loss), etc. [40,41]. The attacker is then attempting to recover a portion/total of the plain text and the key. Besides, the attacker could have decryption key as well as specific conditions but does not have encryption information. In order to break the confidentiality of the proposed technology, along with the key generation technology of the symmetric encryption system, an attacker must acquire a lot of mandatory information that's it:

- A. Main Key
- B. Sub Key
- C. A mathematical method for key generation:
- D. A mathematical method for layer one coding:
- E. Dynamic DNA-coding value
- F. Mathematical sequence table values for the DNA transformations process.
- G. Two separate sports series,
- H. Selected arbitrary positions from this series,
- I. Selected option NCBI Bank Database etc.

8.2.1. Brute force attack

A brute force attack is an attack in which a programmer tries to guess the keys using the space of a potential alternative to the proposed method. Because the space used by the switches is in a multi-block structure of 16-bit blocks, this produces an $n \times 2^{256}$ -bit space where n is the number of blocks. The keys are impossible to measure mathematically. In addition, even though one key is believed, it is impossible to guess the weights of the 2^{256} used to construct the other keys. In addition, it is computationally difficult to assign all probabilities of protein genes to their corresponding 3-mRNA codecs without any unique segmented protein values. The proposed cryptographic scheme is thus resilient to a brute force attack.

8.2.2. Known plain text attack

In a known plain text attack [42], the attacker tries to locate the key by evaluating the relation between the two sets of plain text and the

cypher text. First, the secret is difficult to decipher, and genetic computation requires many large-scale calculations. In this case, even though the attacker retrieves the group key, a new set of keys is generated for each execution. The suggested strategies will also survive the known assault on plain text.

8.2.3. Cipher-text attack only

The attacker will then have access to a collection of cypher scripts as well as some plaintext information, such as the language or statistical distribution of plaintext characters. An attack is assumed to have been successful if the attacker can deduce the clear text and possibly the key this will not function in the proposed method due to the high entropy of the created switches, which are regular XOR text. Additional logical and genetic operations can increase the complexity of the standard text. In other words, the most common cypher text characters would not be assigned to the most frequent cypher text characters. The relationship between regular text and cipher text has become more complicated due to the various regular text and cipher-text (as was observed in all simulated test cases). The proposed cypher scheme is therefore only capable of withstanding the cipher-text attack.

8.2.4. Differential Cipher Analysis attack

Differential Cipher Analysis is a limited plaintext attack in which the attacker extracts cipher text from a collection of chosen standard scripts. This attack also failed due to the latest generation of keys that generated different encrypted texts every time.

8.3. CIA achievement

- **Confidentiality** Means involves unauthorized parties when sharing information. Confidentiality has been generated in the proposed method, where all sent entities and parameters are encrypted.
- **Data integrity** guarantees that the user does not receive any information that has been changed due to insertion, deletion and alteration. In other words, even though the data is tampered with the receiver must provide some of processes to guarantee that information retrieved is updated. The integrity of data was accomplished by segmentation in this scheme. The fractional protein values can vary between the transmitter and the receiver if the attacker maliciously modifies the data before the recipient processes it.
- **Availability:** Availability means ensuring the knowledge reaches users wherever it is needed. The proposed encryption techniques are virtual and are thus useable at all times. It also supports broad formats of text data for encoding and decoding without missing information, i.e. a lossless scheme where no bit is lost during transmission. The proposed encryption scheme algorithm has also been tested using whitespace and special characters for sizes up to 10,000 plaintext characters.

9. Conclusions and future works

With the huge development of data and computing technologies the cloud final users are growing with their personal and professional use. The reason behind such success is their effective and flexible computational and storage solutions based on the applications needs the cloud provides data and computational services on request. There are still many barriers regarding data security and privacy that need to be resolved. Although the cloud facilitates access to data storage with flexibility and ease of management, there are still possibilities for unauthorized attacks and malicious activities. The cloud server may secretly store sensitive data. Therefore, data security is of primary concern. In the proposed work, a novel cryptographic technique based on logical-mathematical functions and genetics techniques to enhance the secrecy level of data while encryption, initially the exploitation of logical operations, such as (XOR, XNOR, shifting) with split the original plaintext and key into equal parts that hold the characteristics of diffusion and confusion cipher that

ensures first –layer secrecy and to increase its privacy, it used Genetics based on the Central Dogma of Molecular Biology for cryptographic purpose through simulation of the natural processes of genetic cryptography (translation from binary to DNA bases), transcription (regeneration from DNA to mRNA), and translation (regeneration from mRNA to protein) to provides second –layer secrecy. Finally, our proposed scheme uses two encryption layers with new key generation technologies leading to higher security.

The result of the proposed algorithm outperforms another genetic cipher (DNA-ElGamal, RC4-DNA-Alg, Abbasy, CryptoGA, Goyal, BDEA, D-GET) and symmetric key cryptographic algorithms (DES, AES, Blowfish, LED, NLCA) in terms of cipher-text size, encryption time, throughput, and security level.

The proposed encryption scheme has been proven safe for brute force, known plain-text attacks, encrypted text only, and differential cryptanalysis attacks. It has been tested on various data, including whitespace and special characters; also, the proposed encryption system fulfills the CIA's principle.

In future work, the proposed algorithm can be improved by reducing the space complexity to meet the memory requirements' challenge. Genetics cryptography is still in a nascent state, and more robust encryption systems can be built using this computational model.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors would like to thank the anonymous reviewers for their comments and suggestions to enhance this manuscript's quality. Also, I need to thank my guide Prof. Dr Sudhir B. Jagtap for his guidance to my work.

References

- [1] F.M. Groom, "The basics of cloud computing," in *Enterprise Cloud Computing for Non-Engineers*, 2018.
- [2] F. Thabit, A. Prof, S. Alhomdy, A.H.A. Al-ahdal, Exploration of Security Challenges in Cloud Computing: Issues , Threats , and Attacks with Their Alleviating Techniques Exploration of Security Challenges in Cloud Computing : Issues , Threats , and Attacks with Their Alleviating Techniques 12th, vol. 10, Journal of Information and Computational Science, USA, 2020, pp. 35–56, 1, December, Submitted for publication.
- [3] M. Zhou, R. Zhang, W. Xie, W. Qian, A. Zhou, Security and Privacy in Cloud Computing: A Survey, 2010, <https://doi.org/10.1109/SKG.2010.19>.
- [4] S. Yi, Z. Qin, Q. Li, Security and privacy issues of fog computing, A survey (2015), https://doi.org/10.1007/978-3-319-21837-3_67.
- [5] M. Sabry, M. Hashem, T. Nazmy, Three reversible data encoding algorithms based on DNA and amino acids' structure, Int. J. Comput. Appl. (2012), <https://doi.org/10.5120/8588-2339>.
- [6] M.R. Biswas, K.M.R. Alam, S. Tamura, Y. Morimoto, A technique for DNA cryptography based on dynamic mechanisms, J. Inf. Secur. Appl. 48 (2019) 102363, <https://doi.org/10.1016/j.jisa.2019.102363>.
- [7] H.M. Mousa, DNA-genetic encryption technique, Int. J. Comput. Netw. Inf. Secur. 8 (7) (2016) 1–9, <https://doi.org/10.5815/ijcnis.2016.07.01>.
- [8] A. Akanksha, A. Bhopale, J. Sharma, A. Meer Shizan, D. Gautam, Implementation of DNA algorithm for secure voice communication, Int. J. Sci. Eng. Res. 3 (6) (2012) 1–5.
- [9] M. Ali, S.U. Khan, A.V. Vasilakos, Security in cloud computing: opportunities and challenges, Inf. Sci. (2015), <https://doi.org/10.1016/j.ins.2015.01.025>.
- [10] N. Kshetri, Privacy and security issues in cloud computing: the role of institutions and institutional evolution, Telecommun. Pol. (2013), <https://doi.org/10.1016/j.tepol.2012.04.011>.
- [11] D. Chen, H. Zhao, Data Security and Privacy Protection Issues in Cloud Computing, 2012, <https://doi.org/10.1109/ICCSEE.2012.193>.
- [12] S. Pearson, A. Benameur, Privacy, Security and Trust Issues Arising from Cloud Computing, 2010, <https://doi.org/10.1109/CloudCom.2010.66>.
- [13] A. Bhardwaj, G.V.B. Subrahmanyam, V. Avasthi, H. Sastry, Security Algorithms for Cloud Computing, 2016, <https://doi.org/10.1016/j.procs.2016.05.215>.

- [14] P. Dixit, A.K. Gupta, M.C. Trivedi, V.K. Yadav, Traditional and hybrid encryption techniques: a survey, in: *Lecture Notes on Data Engineering and Communications Technologies*, 2018.
- [15] S. Chowdhury, S.R. Ghosh, A. Paul, Design and implementation of a novel cryptographic technique for network security using genetic algorithms (gas), *Int. J. Innov. Knowl. Concepts* 119 (7) (2019).
- [16] Naccache David, Reiter Michael, Cryptology and network security, *Network* 9 (10) (2012) 1–256, 9783319268231, 3319268236.
- [17] S.S. Khan, P.R. Tuteja, Security in cloud computing using cryptographic algorithms, *Int. J. Innov. Res. Comput. Commun. Eng.* (2015), <https://doi.org/10.15680/ijirce.2015.0301035>.
- [18] P. N. S. V. Keer, S. I. Ali, “Hybrid approach of cryptographic algorithms in cloud computing.”.
- [19] S. B. J. Fursan Thabit, Alhomdy, “A new lightweight cryptographic algorithm for enhancing data security in cloud,” *Glob. Transitions Proc.*
- [20] A. Kalaivani, B. Ananthi, S. Sangeetha, Enhanced hierarchical attribute based encryption with modular padding for improved public auditing in cloud computing using semantic ontology, *Cluster Comput.* (2019), <https://doi.org/10.1007/s10586-018-2346-1>.
- [21] S.K. Pujari, G. Bhattacharjee, S. Bhoi, A Hybridized Model for Image Encryption through Genetic Algorithm and DNA Sequence, 2018, <https://doi.org/10.1016/j.procs.2017.12.023>.
- [22] C. Chunka, R.S. Goswami, S. Banerjee, A Novel Approach to Generate Symmetric Key in Cryptography Using Genetic Algorithm (GA), 2019, https://doi.org/10.1007/978-981-13-1951-8_64.
- [23] S. Kalsi, H. Kaur, V. Chang, DNA cryptography and deep learning using genetic algorithm with NW algorithm for key generation, *J. Med. Syst.* (2018), <https://doi.org/10.1007/s10916-017-0851-z>.
- [24] M. Cimi Thomas, S. Sheeja, Secure symmetric encryption scheme using genetic algorithm, *Int. J. Appl. Eng. Res.* 12 (21) (2017) 10828–10833.
- [25] M. Sohal, S. Sharma, BDNA-A DNA inspired symmetric key cryptographic technique to secure cloud computing, *J. King Saud Univ. - Comput. Inf. Sci.* (2018), <https://doi.org/10.1016/j.jksuci.2018.09.024>.
- [26] M. Tahir, M. Sardaraz, Z. Mehmood, S. Muhammad, CryptoGA: a cryptosystem based on genetic algorithm for cloud data security, *Cluster Comput.* 4 (2020), <https://doi.org/10.1007/s10586-020-03157-4>.
- [27] A. Aieh, A. Sen, S.R. Dash, S. Dehuri, Deoxyribonucleic Acid (DNA) for a Shared Secret Key Cryptosystem with Diffie Hellman Key Sharing Technique, 2015, <https://doi.org/10.1109/C3IT.2015.7060130>.
- [28] S. Namasudra, D. Devi, S. Kadry, R. Sundarasekar, A. Shanthini, Towards DNA based data security in the cloud computing environment, *Comput. Commun.* 151 (2020) 539–547, <https://doi.org/10.1016/j.comcom.2019.12.041>.
- [29] N.H. UbaidurRahman, C. Balamurugan, R. Mariappan, A novel DNA computing based encryption and decryption algorithm, *Procedia Comput. Sci.* 46 (2014) (2015) 463–475, <https://doi.org/10.1016/j.procs.2015.02.045>.
- [30] M.A. Wright, The advanced encryption standard, *Netw. Secur.* (2001), [https://doi.org/10.1016/S1353-4858\(01\)001018-2](https://doi.org/10.1016/S1353-4858(01)001018-2).
- [31] A.U. Rahman, S.U. Miah, S. Azad, Advanced encryption standard, in: *Practical Cryptography: Algorithms and Implementations Using C++*, 2014.
- [32] M.N. Valmik, P.V.K. Kshirsagar, “Blowfish Algorithm,” *IOSR J. Comput. Eng.* (2014), <https://doi.org/10.9790/0661-162108083>.
- [33] G. Bansod, N. Raval, N. Pisharoty, Implementation of a new lightweight encryption design for embedded security, *IEEE Trans. Inf. Forensics Secur.* (2015), <https://doi.org/10.1109/TIFS.2014.2365734>.
- [34] M. Thangavel, P. Varalakshmi, Enhanced DNA and elgamal cryptosystem for secure data storage and retrieval in cloud, *Cluster Comput.* 21 (2) (2018), <https://doi.org/10.1007/s10586-017-1368-4>.
- [35] S. Narendren, Y.B. Yathish, C.M. B. A cryptosystem using two layers of security - DNA and RSA cryptography 119 (7) (2018) 217–224.
- [36] R.K. Ahmed, I.J. Mohammed, Developing a new hybrid cipher algorithm using DNA and RC4, *Int. J. Adv. Comput. Sci. Appl.* 8 (10) (2017) 171–176. Available: www.ijacsa.thesai.org.
- [37] S. Goyal, S. Jain, A secure cryptographic cloud communication using DNA cryptographic technique, *Proc. Int. Conf. Inven. Comput. Technol. ICICT 2016* (2016) 2016, <https://doi.org/10.1109/INVENTIVE.2016.7830158>.
- [38] S. Basu, M. Karupiah, M. Nasipuri, A.K. Halder, N. Radhakrishnan, Bio-inspired cryptosystem with DNA cryptography and neural networks, *J. Syst. Architect.* 94 (September 2018) (2019) 24–31, <https://doi.org/10.1016/j.sysarc.2019.02.005>.
- [39] C.E. Shannon, A mathematical theory of communication, *Bell Syst. Tech. J.* (1948), <https://doi.org/10.1002/j.1538-7305.1948.tb01338.x>.
- [40] A. Fatima, N.K. Nishchal, Discussion on comparative analysis and a new attack on optical asymmetric cryptosystem, *J. Opt. Soc. Am. A* (2016), <https://doi.org/10.1364/josaa.33.002034>.
- [41] H. Chen, C. Tanougast, Z. Liu, B. Hao, Securing color image by using hyperchaotic system in gyrator transform domains, *Opt. Quant. Electron.* (2016), <https://doi.org/10.1007/s11082-016-0669-9>.
- [42] S.K. Rajput, N.K. Nishchal, Known-plaintext attack on encryption domain independent optical asymmetric cryptosystem, *Opt Commun.* (2013), <https://doi.org/10.1016/j.optcom.2013.06.036>.